1 Notes: Wagner's model

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Free surface predicted by Wagner' theory

$$\eta = \frac{2y\tan\beta}{\pi}\arcsin\left(\frac{\pi W_0 t}{2y\tan\beta}\right) - W_0 t \tag{1}$$

Wet length by Wagner's theory

$$c(t) = \frac{\pi W_0 t}{2\tan\beta} \tag{2}$$

Vertical position of spray root by Wagner's theory

$$z_c(t) = \left(\frac{\pi}{2} - 1\right) W_0 t \tag{3}$$

Trajectory of spray root by Wagner's theory

$$z_c = \left(1 - \frac{2}{\pi}\right)\tan\beta y \tag{4}$$

Vertical coordinate of the plate bottom surface

$$z_p(y) = \tan\beta y - W_0 t \tag{5}$$

Horizontal position of geometrical intersection line of plate and mean water surface ($z_p = 0$)

$$y_i = \frac{W_0 t}{\tan\beta} \tag{6}$$

"Thickness" of the root region

$$d = c - y_i = \left(\frac{\pi}{2} - 1\right)\frac{W_0 t}{\tan\beta} \tag{7}$$

In general, if define γ as the angle between the root trajectory and the horizontal direction, the relation between γ and β is a function of root speed V_s and plate speed W_0

$$\frac{V_s}{W_0} = \frac{1}{\cos\gamma\tan\beta - \sin\gamma} \tag{8}$$

In Wagner's theory, the root speed is

$$V_s = \sqrt{\dot{c}^2 + \dot{z}_c^2} = \sqrt{\left(\frac{\pi}{2\tan\beta}\right)^2 + \left(\frac{\pi}{2} - 1\right)^2} W_0 \tag{9}$$

Therefore, in Wagner's theory, V_s/W_0 is a function of deadrise angle β. The rate of change of wet length

$$\dot{c} = \frac{\pi W_0}{2\tan\beta} \tag{10}$$

Therefore, the ratio $\dot{c}/W_0$ is $\pi/(2\tan\beta)$. Therefore, in Wagner's theory, V_s/W_0 is not proportional to $\dot{c}/W_0$ for all deadrise angle β, as reported in Kanso's paper.

Pressure distribution on the wedge (Faltinsen, 2005, pp. 307)

$$p = -\rho\frac{\partial\phi}{\partial t} = \rho\frac{dW_0}{dt}\sqrt{c^2-x^2} + \rho W_0\frac{c}{\sqrt{c^2-x^2}}\frac{dc}{dt} \tag{11}$$

If wedge moves at constant speed $W_0 = \text{const}$,

$$p = \rho W_0\frac{c}{\sqrt{c^2-x^2}}\frac{dc}{dt} \tag{12}$$

Integrate to get the force

$$F = \int_{-c}^{c} p dx = \int_{-c}^{c}\rho W_0\frac{c}{\sqrt{c^2-x^2}}\frac{dc}{dt}dx = \rho W_0 c\frac{dc}{dt}\int_{-c}^{c}\frac{1}{\sqrt{c^2-x^2}}dx$$

$$F = \rho W_0 c\frac{dc}{dt}\arcsin\left(\frac{x}{c}\right)\Big|_{-c}^{c} = \rho W_0 c\frac{dc}{dt}\pi$$

Find the equivalent point of application of F on a half-wedge (plate)

$$\int_0^c pxdx = F_p L_F = 0.5FL_F$$

$$\int_0^c pxdx = \int_0^c \rho W_0\frac{cx}{\sqrt{c^2-x^2}}\frac{dc}{dt}dx = \rho W_0 c\frac{dc}{dt}\int_0^c\frac{x}{\sqrt{c^2-x^2}}dx$$

$$\int_0^c pxdx = \rho W_0 c\frac{dc}{dt}\left[-\sqrt{c^2-x^2}\right]\Big|_0^c = \rho W_0 c^2\frac{dc}{dt} = 0.5FL_F$$

$$\rho W_0 c^2\frac{dc}{dt} = 0.5\rho W_0 c\frac{dc}{dt}\pi L_F$$

Thus, distance from the vertex to the the equivalent point of application of F along the wedge is

$$L_F = \frac{2}{\pi}c = \frac{2}{\pi}\frac{\pi W_0 t}{2\tan\beta} = \frac{W_0 t}{\tan\beta} \tag{13}$$

Total normal force on the wedge (Faltinsen, 2005, pp. 307)

$$F = \rho\pi W_0 c\frac{dc}{dt} + \rho\frac{\pi}{2}c^2\frac{dW_0}{dt} \tag{14}$$

If W_0 is constant over time, the total force (per unit width) is

$$F = \rho\pi W_0 c\frac{dc}{dt} \tag{15}$$

Because

$$c = \frac{\pi W_0 t}{2\tan\beta}$$

Thus, the total force

$$F = \rho\pi W_0 \frac{\pi W_0 t}{2\tan\beta}\frac{\pi W_0}{2\tan\beta} = \frac{\pi^3 \rho W_0^3 t}{4\tan^2\beta} \tag{16}$$

The time it takes for the root to reach the trailing edge is

$$t_r = \frac{B\cos\beta}{dc/dt} = \frac{B\cos\beta}{\frac{\pi W_0}{2\tan\beta}} = \frac{2B\sin\beta}{\pi W_0} \tag{17}$$

At $t = t_r$, the total force reaches the maximum value

$$F_m = \frac{\pi^3 \rho W_0^3}{4\tan^2\beta}\frac{2B\sin\beta}{\pi W_0} = \frac{\pi^2\cos^2\beta}{2\sin\beta}\rho B W_0^2 \tag{18}$$

F_m increases as β decreases. The pressure distribution is not unifrom, however, if we apply the total force at the center of the beam, it provides an overprediction of the maximum deflection. The deflection is calculated by (Mark's Standard Handbook for Mechanical Engineers, pp. 5-23, Table 5.2.2)

$$\delta_m = \frac{F_m l B'^3}{192EI} \tag{19}$$

where B' is the length of the 8020 transverse beam between the two clamped ends, i.e., $B' = 0.279$ m and l is the width of the 8020 beam, i.e., $l = 3$ in $=$ 0.0762 m. The Young's modulus of T6061 Aluminum is $E = 68.9$ GPa and the moment of inertia of 8020-1530 beam is $I = 0.4824$ in^4 $= 2.0079 \times 10^{-7}$ m^4. Note, for our experiment, only half a wedge is under impact, so the force should be half of F_m.

$$F_{pm} = 0.5F_m = \frac{\pi^2\cos^2\beta}{4\sin\beta}\rho B W_0^2 \tag{20}$$